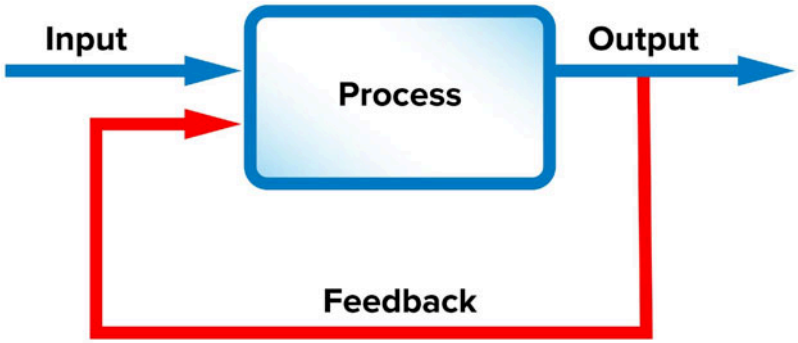
Homeostasis, Feedback Loops and Signalling

Human Biology 2026



Dr Shanker Karunanithi
School of Medical Sciences
Faculty of Medicine and Health

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Acknowledgement of country

We recognise and pay respect to the Elders and communities – past, present, and emerging – of the lands that the University of Sydney's campuses stand on. For thousands of years they have shared and exchanged knowledges across innumerable generations for the benefit of all.

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Major Concepts

How do we maintain stability in the body?

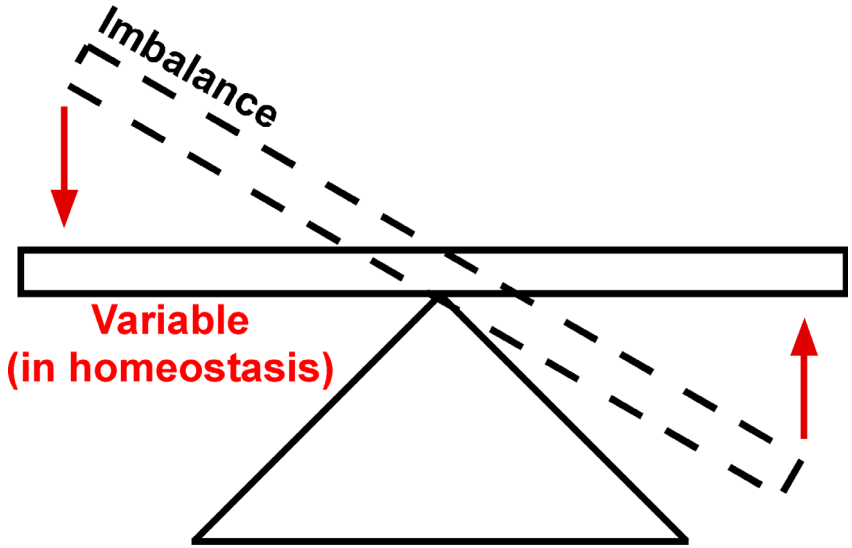
Feedback loops – positive and negative

How do cells/tissues/organs communicate with themselves and the outside world?

HOMEOSTASIS

Homeostasis

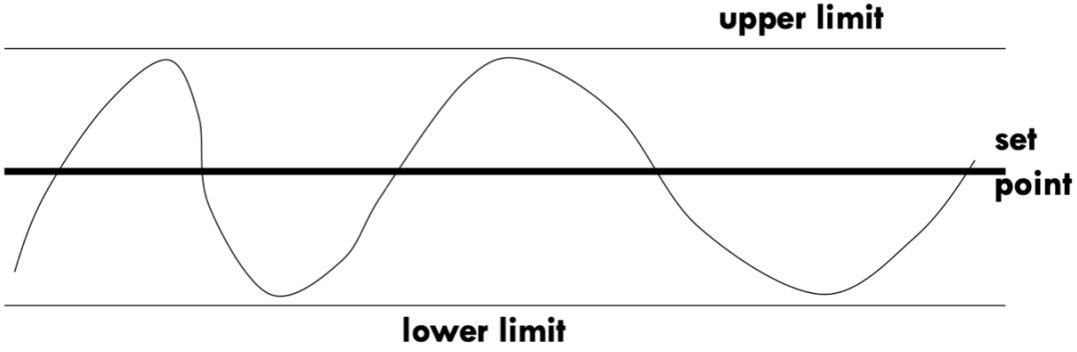
Walter B. Cannon



Maintaining stability within the internal environment

Homeostasis

Homeostasis is maintained through mechanisms that counteract deviations, primarily by using negative feedback



Examples of homeostatically maintained variables within the body:

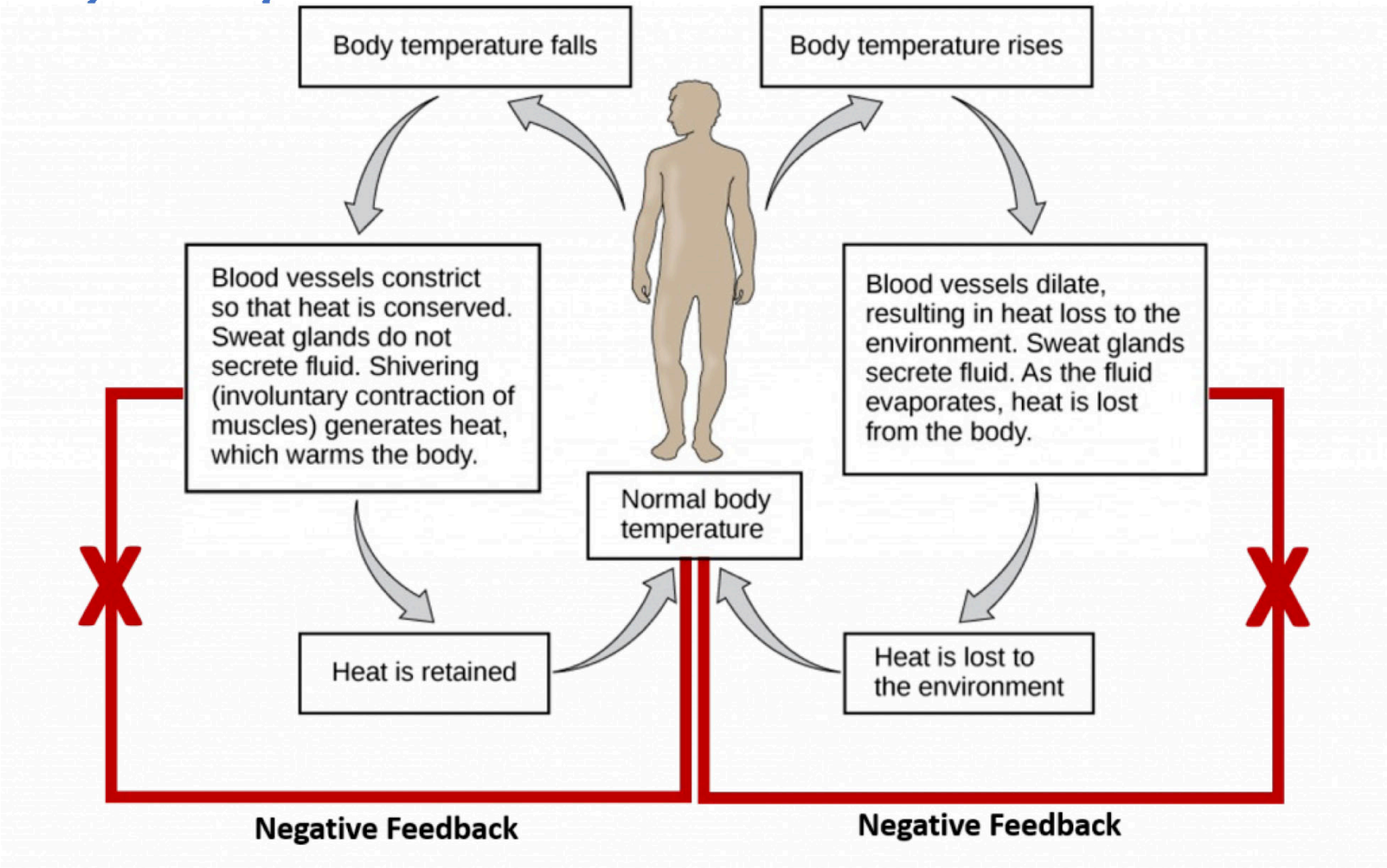
1) body temperature (set point ~37 °C ; regulated by the nervous system)
2) blood glucose (set point ~5 mM; regulated by the endocrine system)
3) blood pH (set point ~7.4; regulated by the respiratory and urinary systems)

Feedback Loops

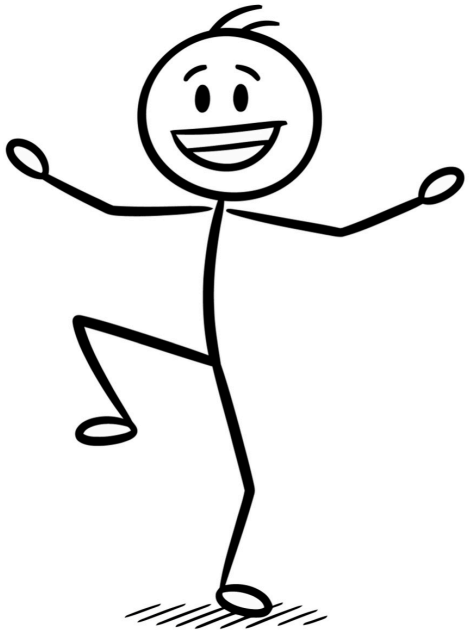
- **Negative** feedback loops act to reduce deviations of controlled variables from their set points, restoring them toward equilibrium

- **Positive** feedback loops act to amplify deviations of controlled variables from their set points, driving them further away from equilibrium. Examples being childbirth and blood clotting.

Body temperature maintenance

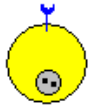


Practical demonstration of homeostasis

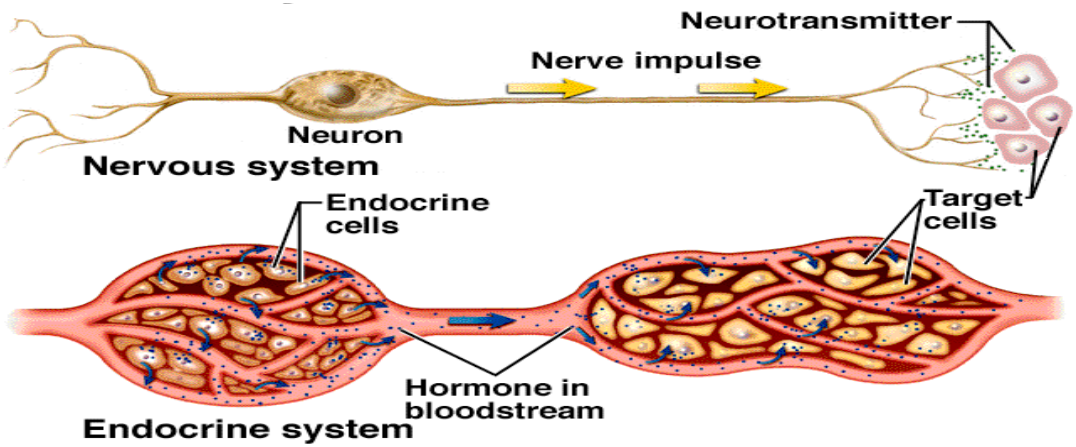


In order to maintain homeostasis, we need systems that can control our entire body

The nervous system sends very **fast** electric messages to our body



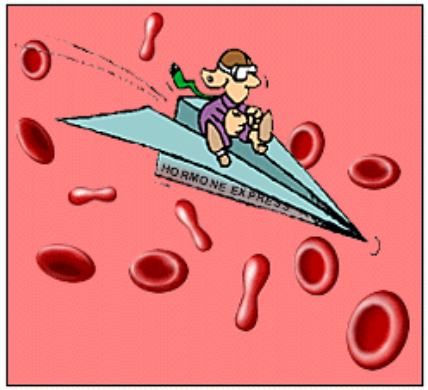
A **slower** acting chemical based messaging system that helps maintain **homeostasis** – the **endocrine** system



SUMMARY	Nervous system	Endocrine system
Anatomical arrangement	wired	wireless
Chemical messenger	neurotransmitter	hormone
Speed of response	milliseconds	minutes or longer
Duration of action	milliseconds	minutes, hours or days
Function	Rapid, precise, short-acting	Longer duration, effects may be marked and widespread in the body

Endocrine system

- In this system, a variety of organs throughout our body release chemicals into our bloodstream. When these chemical messengers arrive at cells in our body – they tell those cells how to behave and act



A hormone is a chemical/organic molecule messenger made by endocrine cells

https://youtu.be/Og9XKqZNJ3E

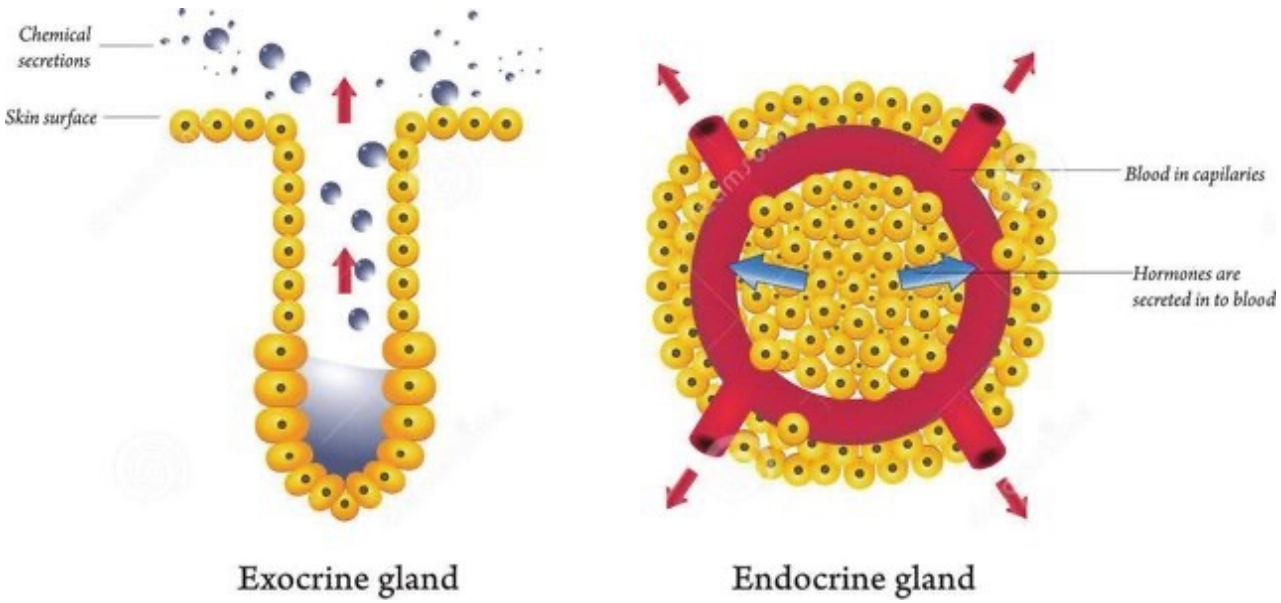
What is the endocrine system?

Endocrine – an internal secretion that pertains to a gland that secretes directly into the bloodstream (e.g. insulin)

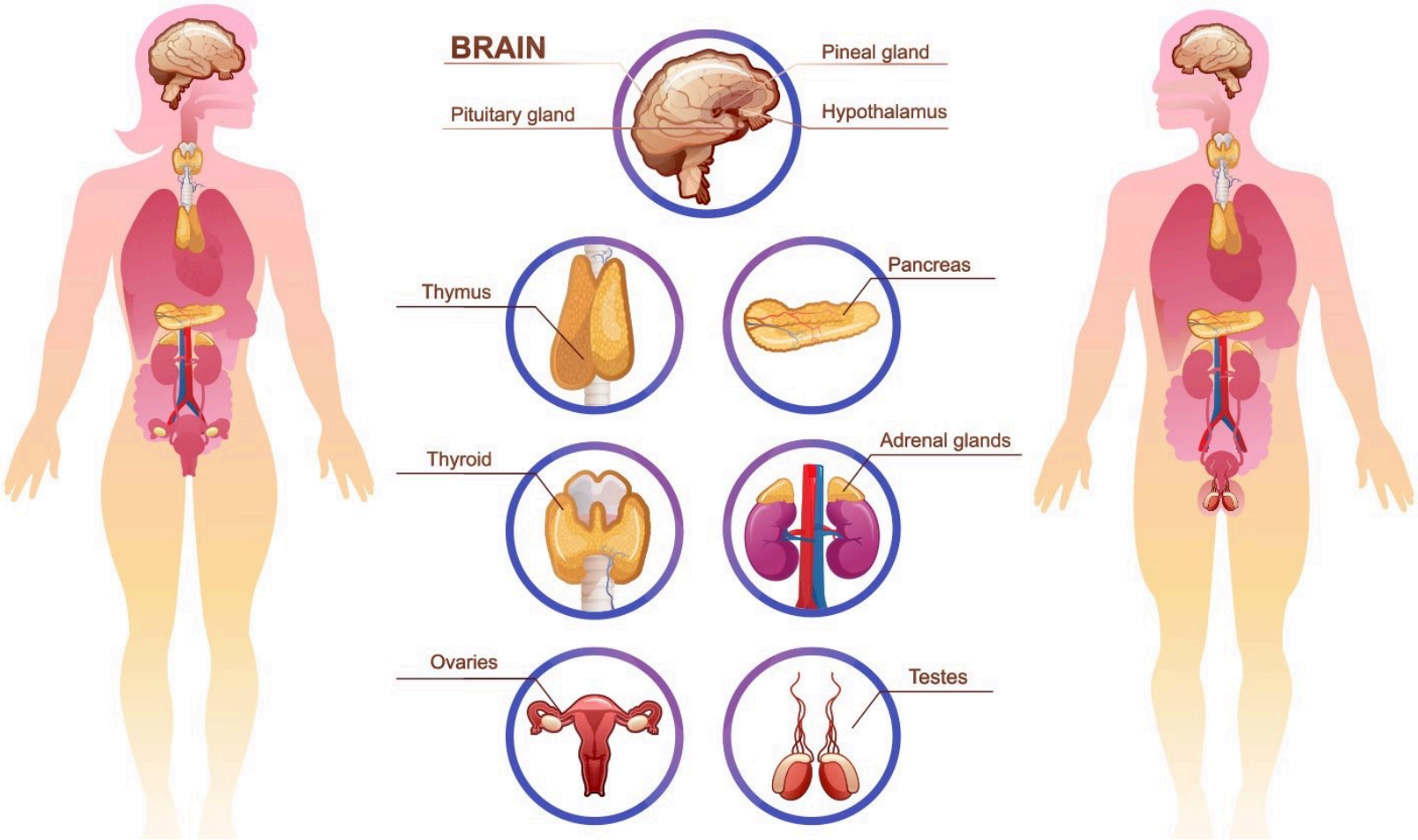
Endocrine gland – a ductless gland that produces an internal secretion discharged into the blood or lymph and circulated to all parts of the body

Exocrine – external secretion via ducts to epithelial surface (e.g. sweat)

If it is circulated to all parts of the body – how can there be specific actions???



ENDOCRINE SYSTEM



https://www.endocrine.org/patient-engagement

Functions of hormones

- **Reproduction, growth and development**

  Sex Steroids, thyroid hormones, prolactin, growth hormone

- **Maintenance of internal environment**

  Aldosterone, parathyroid hormone, vitamin D

- **Energy production, utilization and storage**

  Insulin, glucagon, thyroid hormones, cortisol, growth hormone

CELL SIGNALLING

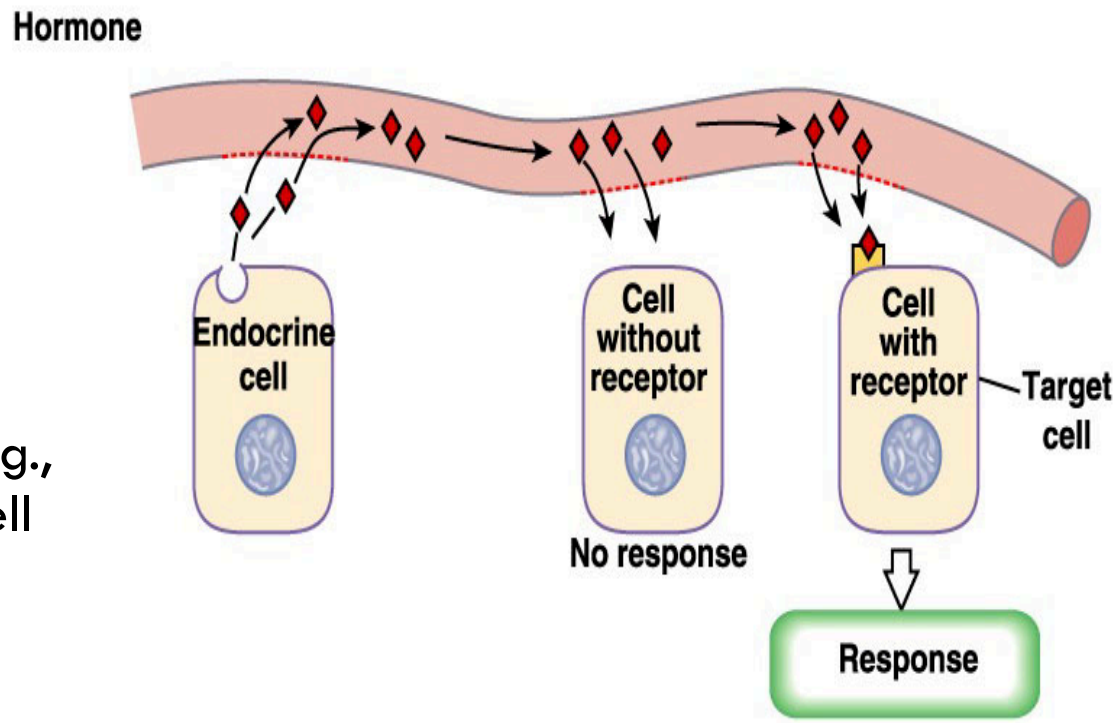
CELL-TO-CELL SIGNALLING

Signalling

We will investigate cell signalling using the example of hormonal signalling

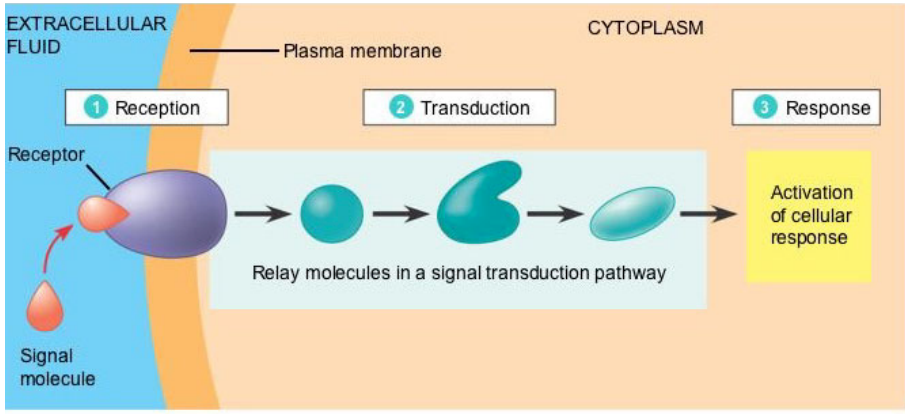
How do hormones (like insulin) work?

- High affinity interaction with **ligand** (molecules which can bind to a receptor, e.g., hormones) binding to **receptor** on target cell
- High specificity for ligand
- Receptor has specific distribution patterns within body
- Hormones target organs/cells expressing the specific receptors

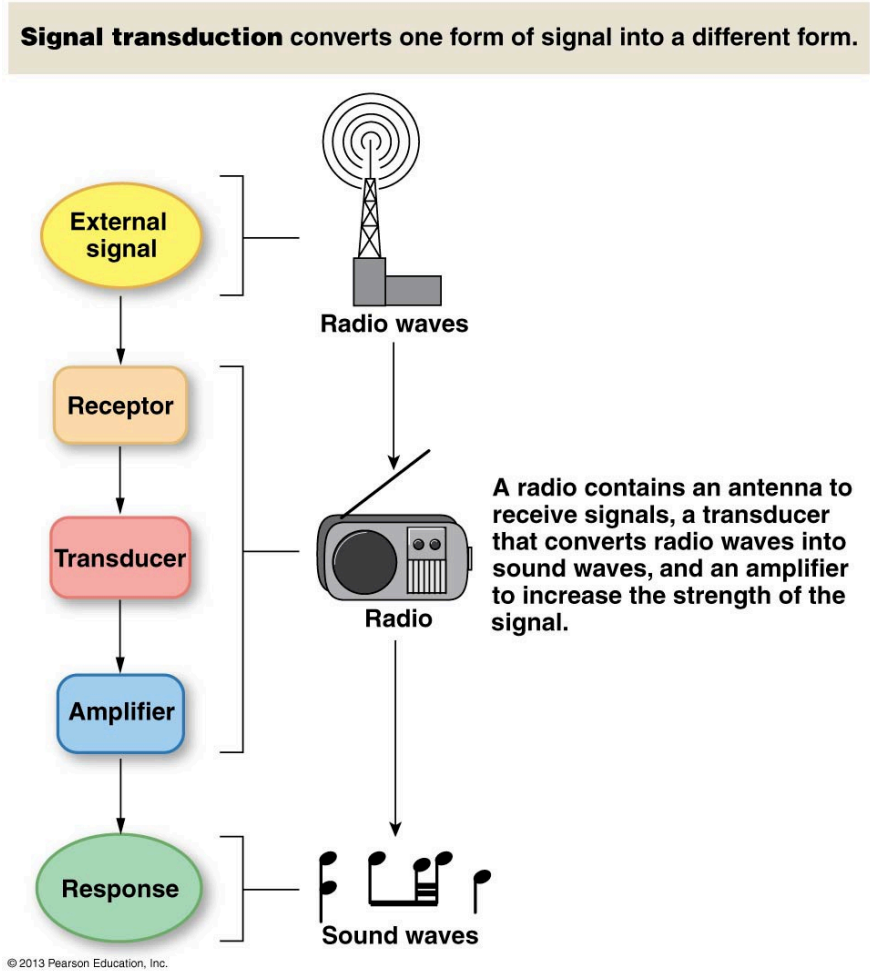


INTRACELLULAR SIGNALLING: Signal transduction

Signal transduction



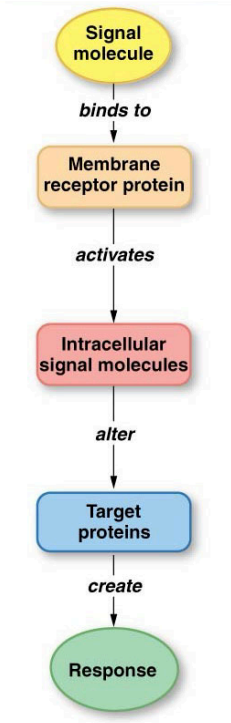
- In response to an external stimulus (e.g. noradrenaline)
- Signal leads to cell response
- Cellular responses could be to: secrete, contract, move, divide, die, survive, kill, differentiate, etc.

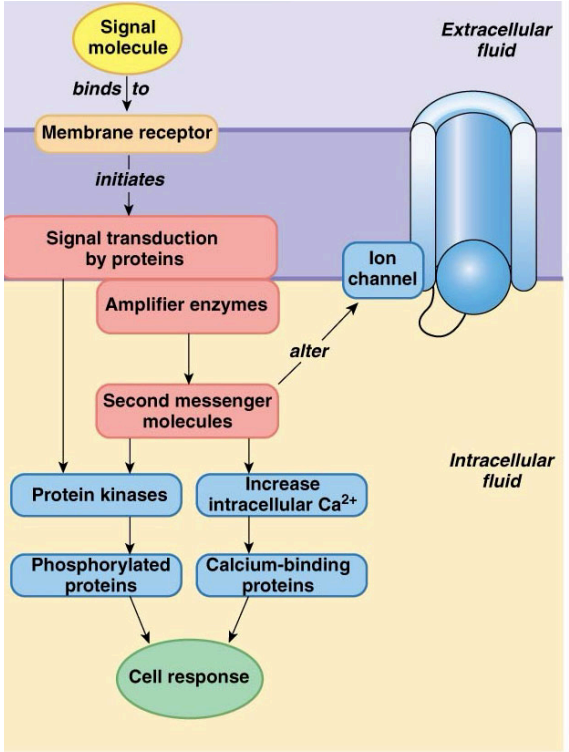


Signal transduction pathways

Pathways:

- Convert extracellular signals into intracellular responses
- Use receptors, transducers, and second messengers
- Amplify small signals into large cellular effects
- Activate specific target proteins to produce a response





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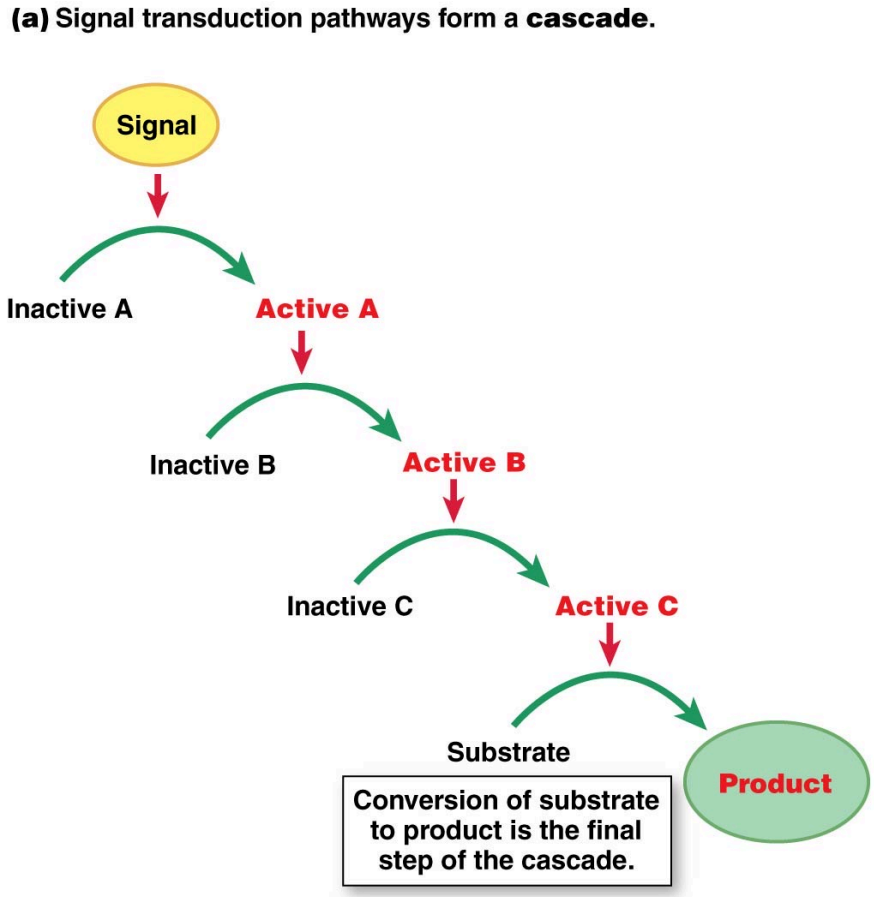
Signalling cascade



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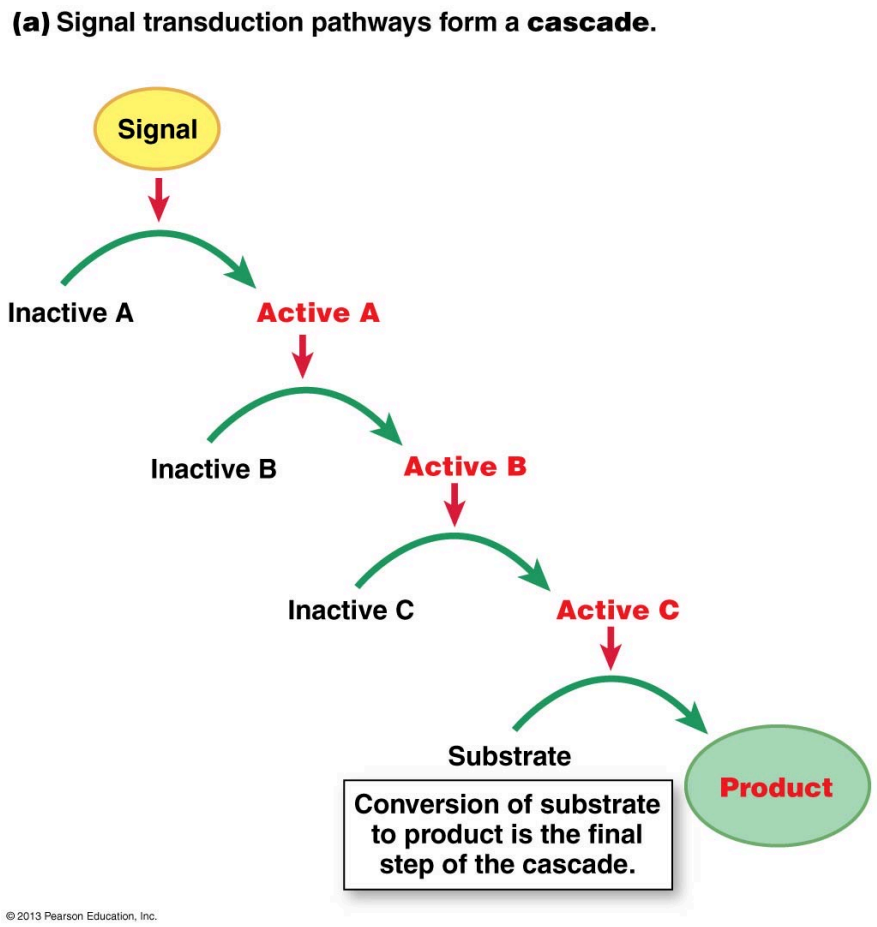
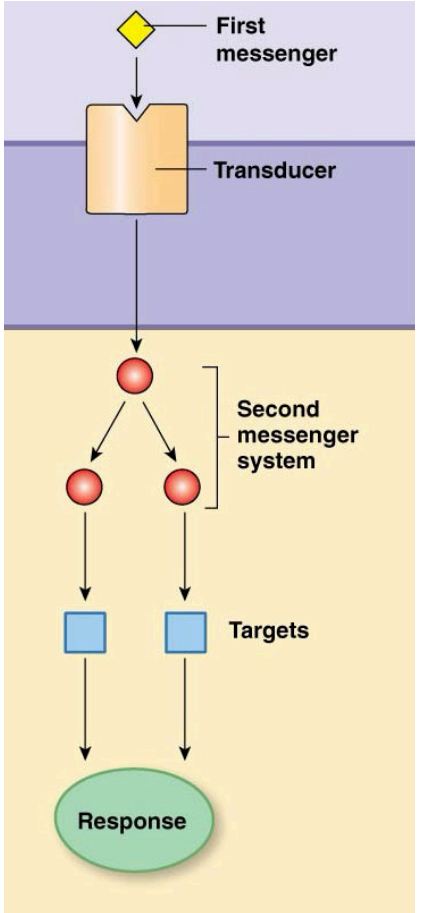
While signal transduction is the overall pathway, the cascade represents the step-by-step activation inside the cell. Features of the cascade:

- One activated molecule activates the next in a step-by-step manner.
- Each step depends on the previous one.
- Signal flow is in one direction.
- The final activated molecule produces the cellular response.
- The signal does not directly cause the response — it is relayed through a sequence of intracellular proteins, like **falling dominos** where one triggers the next.

Linking the signalling cascade to the transduction pathway

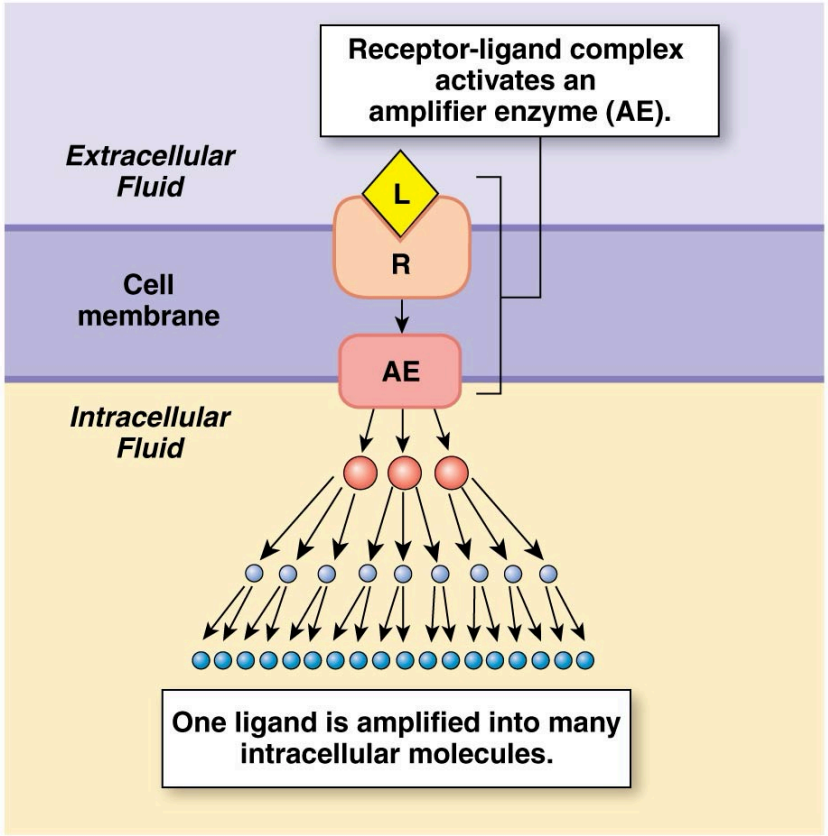




Cascade Diagram (Left)	Signal Transduction Diagram (Right)	Conceptual Meaning
Signal	First messenger	Extracellular signalling molecule
Inactive A → Active A	Transducer (receptor activation)	First intracellular signalling step
Inactive B → Active B	Second messenger system	Amplification via intracellular signalling molecules
Inactive C → Active C	Targets	Activation of downstream effector proteins
Substrate → Product	Response	Final cellular outcome

Signal amplification

(b) Signal amplification allows a small amount of signal to have a large effect.



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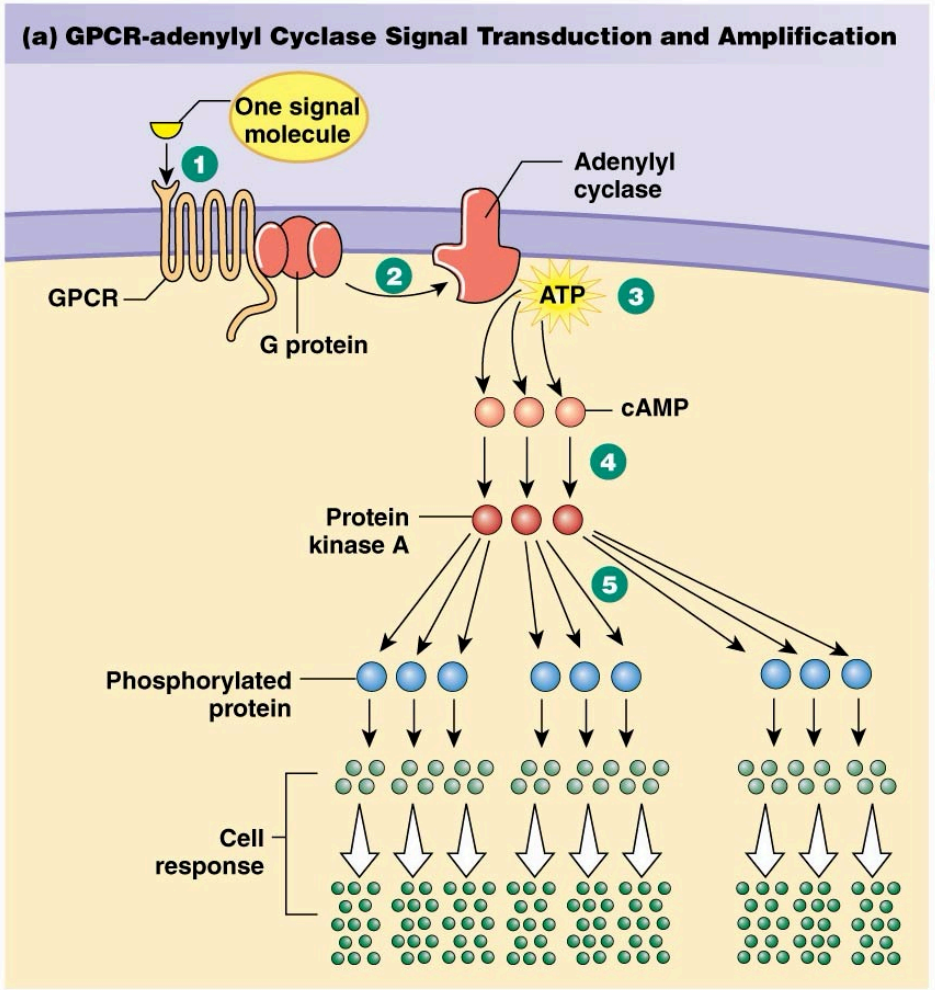
Signal amplification:

- A small concentration of ligand produces a large intracellular response.
- One ligand binds one receptor.
- The receptor activates an amplifier enzyme.
- The amplifier enzyme generates many intracellular signalling molecules.
- Each step increases the strength of the signal.
- Amplification allows cells to respond strongly to small concentrations of ligand.

Water-soluble hormones: signal transduction pathway



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1. Signal molecule binds to G protein–coupled receptor (GPCR), which activates the G protein.
2. G protein turns on adenylyl cyclase, an amplifier enzyme.
3. Adenylyl cyclase converts ATP to cyclic AMP.
4. cAMP activates protein kinase A.
5. Protein kinase A phosphorylates other proteins, leading ultimately to a cellular response.

Hormones that act via cAMP mechanisms:

Epinephrine	Glucagon
ACTH	PTH
FSH	TSH
LH	Calcitonin

Water soluble hormones: peptide hormones, like glucagon (acts on liver cells to ↑blood glucose), can't cross membrane

Require transmembrane receptors

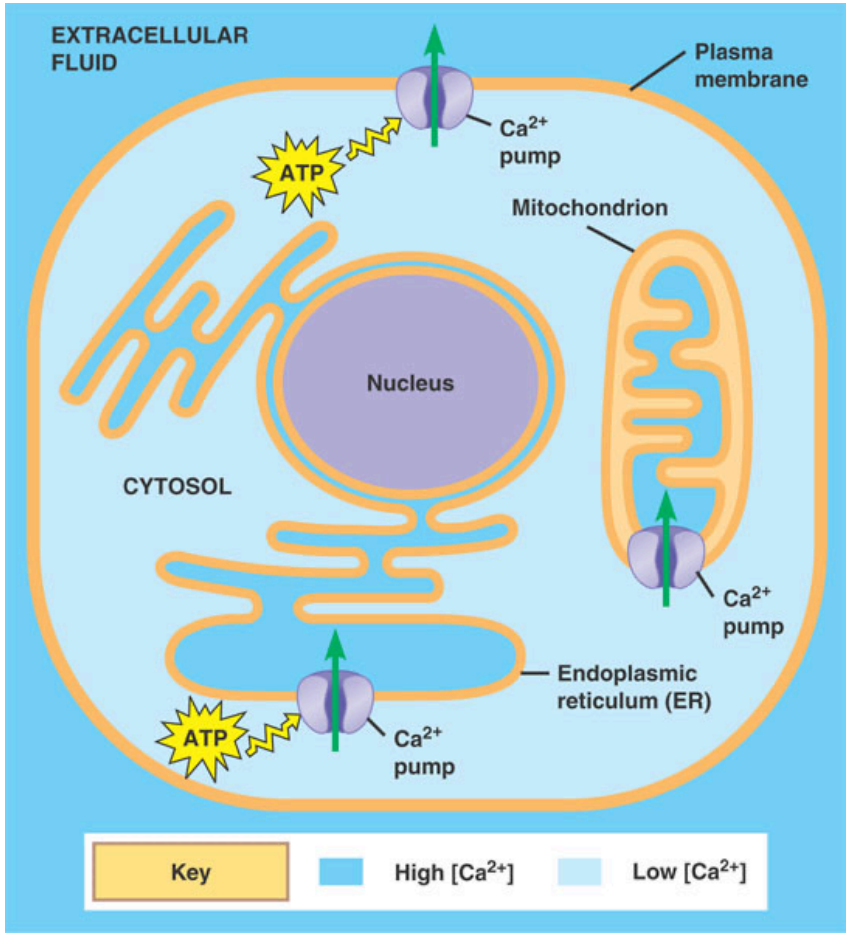
Receptor then transduces signal. **How?**

https://www.youtube.com/watch?v=Nt2r5R0ZO5U

SUMMARY: GPCR Pathway → Signal Transduction Terminology

Specific G protein-coupled receptor pathway	Terminology	Explanation
Hormone / ligand (1st messenger)	Signal molecule	Extracellular signal that initiates the pathway
Receptor (GPCR)	Membrane receptor	Detects the signal and activates intracellular signalling
G protein	Signal transduction protein	Relays signal from receptor to effector enzyme
Adenylyl cyclase	Amplifier enzyme	Produces many second messenger molecules from one signal
cAMP	Second messenger	Intracellular molecule that spreads and amplifies the signal
Protein kinase A (PKA)	Protein kinases	Activated by cAMP to phosphorylate target proteins
Phosphorylated proteins	Phosphorylated proteins	Proteins whose activity has been modified
Ion channels / enzymes / secretion	Cell response mechanisms	Effectors that produce the functional response
Cellular response	Cell response	Final physiological outcome in the target cell

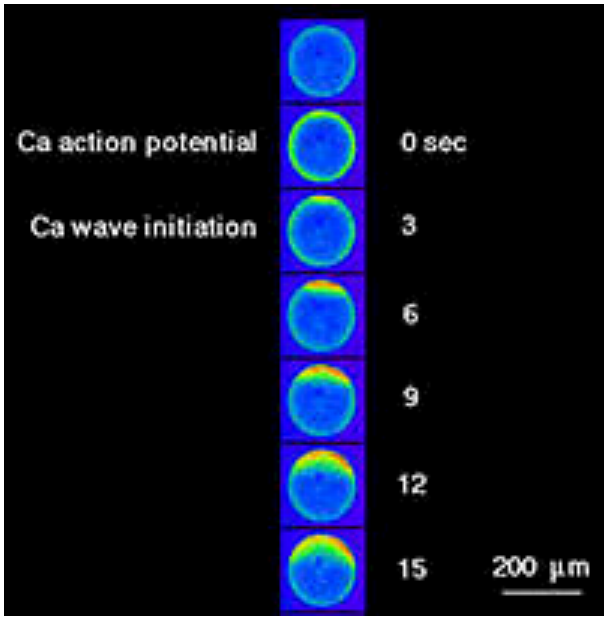
Calcium as a 2nd messenger

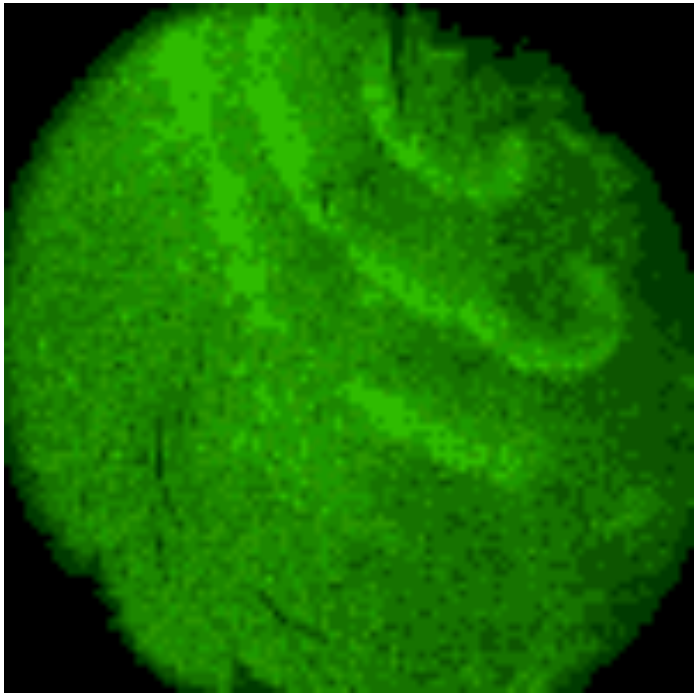


- Resting cytosolic [Ca^{2+}] is extremely low (~10^{-7} M).
- Extracellular and ER [Ca^{2+}] is high (~10^{-3} M) → ~10,000-fold gradient.
- Small increases in Ca^{2+} act as a powerful "call to action".
- Ca^{2+} binds proteins → changes conformation and activity.
- Rapid Ca^{2+} entry or ER release triggers cellular responses.
- Example: Ca^{2+} triggers synaptic vesicle exocytosis.

Calcium waves upon fertilization

Slower mobilisation of calcium





SUMMARY: Examples of 2nd messengers & amplifier enzymes

(c) Second messenger pathways

SECOND MESSENGER	MADE FROM	AMPLIFIER ENZYME	LINKED TO	ACTION	EFFECTS
Nucleotides					
cAMP	ATP	Adenylyl cyclase (membrane)	GPCR*	Activates protein kinases, especially PKA. Binds to ion channels.	Phosphorylates proteins. Alters channel opening.
cGMP	GTP	Guanylyl cyclase (membrane)	Receptor-enzyme	Activates protein kinases, especially PKG.	Phosphorylates proteins.
		Guanylyl cyclase (cytosol)	Nitric oxide (NO)	Binds to ion channels.	Alters channel opening.
Lipid-derived*					
IP_3	Membrane phospholipids	Phospholipase C (membrane)	GPCR	Releases Ca^{2+} from intracellular stores.	See Ca^{2+} effects below.
DAG				Activates protein kinase C.	Phosphorylates proteins.
Ions					
Ca^{2+}				Binds to calmodulin. Binds to other proteins.	Alters enzyme activity. Exocytosis, muscle contraction, cyto-skeleton movement, channel opening.

*GPCR = G protein–coupled receptor. IP_3 = Inositol trisphosphate. DAG = diacylglycerol

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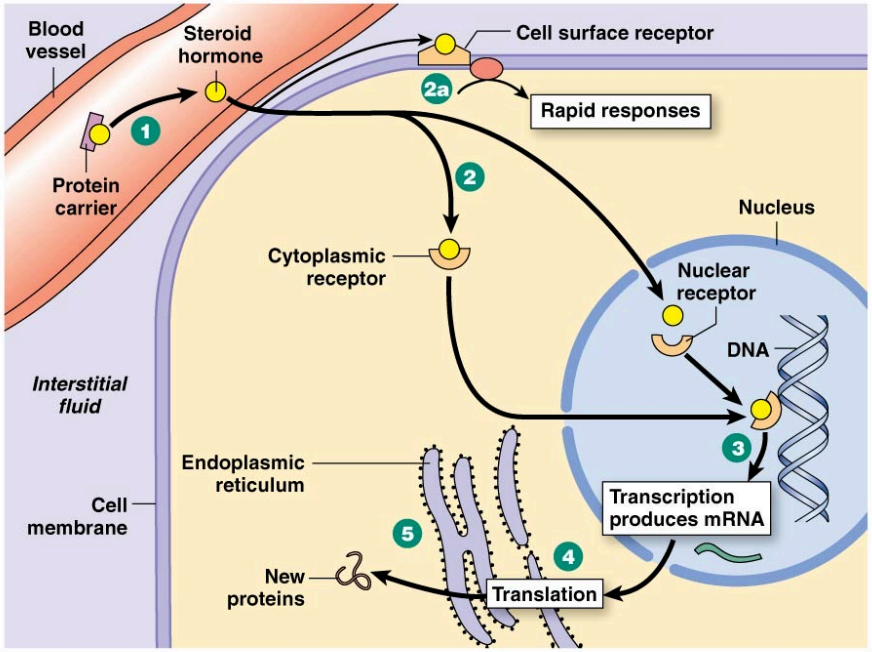
Steroid hormones

(b) Steroid hormones act primarily on intracellular receptors.



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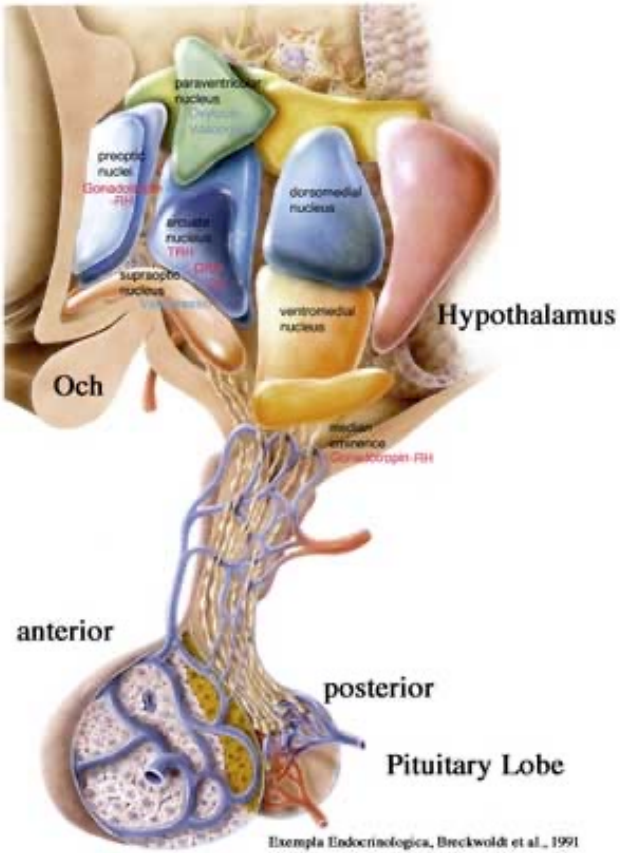
1. Most hydrophobic steroids are bound to plasma protein carriers. Only unbound hormones can diffuse into the target cell.
2. Steroid hormone receptors are in the cytoplasm or nucleus.
2a. Some steroid hormones also bind to membrane receptors that use second messenger systems to create rapid cellular responses.
3. The receptor-hormone complex binds to DNA and activates or represses one or more genes.
4. Activated genes create new mRNA that moves back to the cytoplasm.
5. Translation produces new proteins for cell processes.

Steroid hormones:

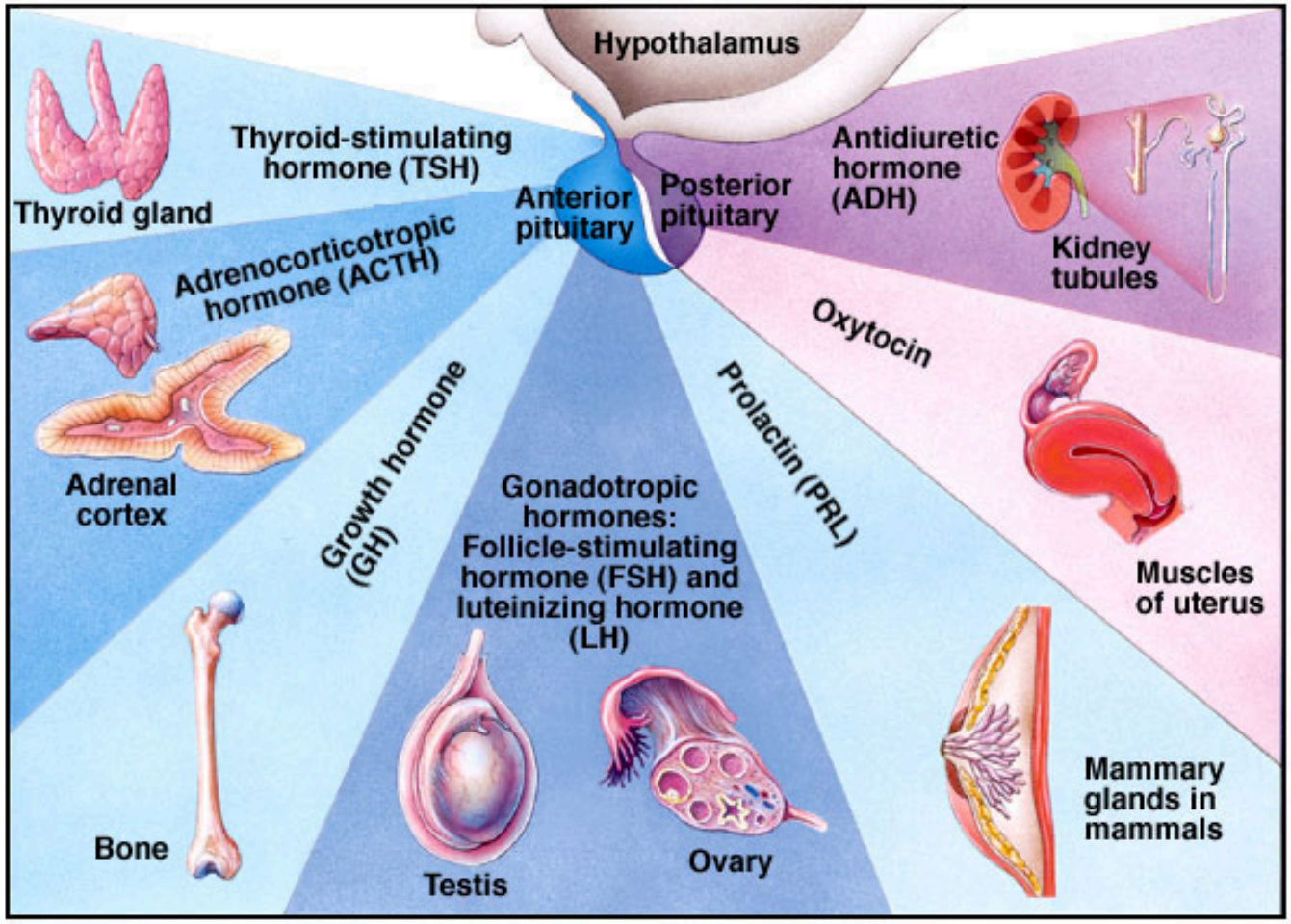
Diffuse across membrane

- Receptor = DNA-binding regulator
- Changes gene expression
- New proteins drive the response
- Slow but sustained effects
- Clinical example: corticosteroids reduce inflammation by suppressing inflammatory gene expression.

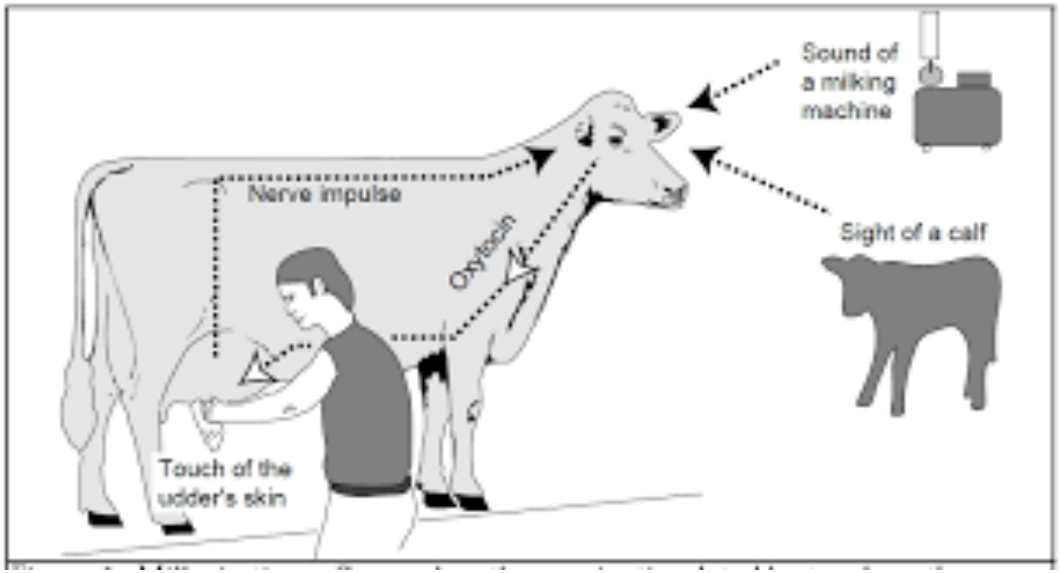
The Master Gland – the Pituitary



Pituitary Gland Hormones



Oxytocin


Sound of
a milking
machine
Nerve impulse
Oxytocin
Sight of a calf
Touch of the
udder's skin

Learning Objectives

Understand the role of feedback loops in maintaining homeostasis – positive and negative

Describe how cells/tissues/organs communicate with themselves and the outside world

Identify the major intracellular signals, how they are generated, and what they do

Appreciate the role of the pituitary gland and pituitary hormones in regulating different organs

Useful References

- Silverthorn, D. U. (2019). Human Physiology: An Integrated Approach, 8th edition. Pearson Education.
- Marieb, E. N., & Hoehn, K. (2019). Human Anatomy & Physiology, 11th edition. Pearson Education.